MAT 3103: Computational Statistics and Probability **Semester: Fall 2019-20**

Lecture 5: Continuous Probability Distribution

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In this chapter we will discuss about some continuous probability distributions. Among all of the continuous probability distributions, Normal distribution, Exponential distribution and Rayleigh distribution are important.

5.1 Normal Distribution

Let, X be a continuous random variable. This X is called normal variable if its probability density function is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2}; -\infty < x < \infty$$

where $E(X) = \mu$ and $V(X) = \sigma^2$. It is usually written as $X \sim N(\mu, \sigma^2)$.

Examples: Age, height, weight, time, lifetime of light bulbs etc; if these are observed from random experiment.

5.1.1 Characteristics of Normal Distribution

- The normal curve is bell-shaped.
- If $X \sim N(\mu, \sigma^2)$, then $Z = \frac{x-\mu}{\sigma} \sim N(0, 1)$. This Z is called standard normal variable.
- The normal curve is symmetric and mesokurtic.
- 95% values of normal variable fall in the limit -2 to $+2$.

5.1.2 Mean & variance calculation using generating functions

The moment generating function(mgf) is

$$\begin{aligned} M_x(t) &= E[e^{tX}] \\ &= \int_{-\infty}^{\infty} e^{tx} f(x)dx \\ &= \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2} dx \\ &= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2} dx \end{aligned}$$

Let, $z = \frac{x-\mu}{\sigma}; -\infty < z < \infty$ Then, $x = \mu + \sigma z$ and $dx = \sigma dz$.

$$\begin{aligned} M_x(t) &= \frac{\sigma}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{t(\mu+\sigma z)} e^{-\frac{1}{2}z^2} dz \\ &= \frac{e^{t\mu}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}[z^2 - 2t\sigma z + t^2\sigma^2]} e^{\frac{1}{2}t^2\sigma^2} dz \\ &= \frac{\mu t + \frac{1}{2}t^2\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z-t\sigma)^2} dz \end{aligned}$$

Let, $u = z - t\sigma$, $du = dz; -\infty < u < \infty$, Then

$$M_x(t) = e^{\mu t + \frac{1}{2}t^2\sigma^2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}u^2} du$$

If $z = \frac{x-\mu}{\sigma}$, then $\int_{-\infty}^{\infty} f(x)dx = \frac{\sigma}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}z^2} dz = 1$

So, $M_x(t) = e^{\mu t + \frac{1}{2}t^2\sigma^2}$

$K_x(t) = \ln M_x(t) = \ln e^{\mu t + \frac{1}{2}t^2\sigma^2} = \mu t + \frac{1}{2}t^2\sigma^2$

K_r =Coefficient of $\frac{t^r}{r!}$ in $K_x(t)$

$K_1 = \mu_1' = \mu$ =Mean
$K_2 = \mu_2 = \sigma^2$ =Variance

Hence, $X \sim N(\mu, \sigma^2)$.

5.1.3 Examples of Normal Distribution

Example 1: The computer failure time X is assumed to follow normal distribution with mean=0.5 hour and standard deviation=0.75 hour. Find the probability that in a day the computer fails to work for
(i)2 hours or more
(ii)less than or equal to 2.5 hours.

Solution: Here, mean(μ) = 0.5 and standard deviation(σ) = 0.75.

We have, $z = \frac{X-\mu}{\sigma}$.

(i) Probability that the computer fails to work for 2 hours or more is,

$$\begin{aligned} P(X \geq 2) &= P(\frac{x-\mu}{\sigma} \geq \frac{2-0.5}{o.75}) \\ &= P(Z \geq 2) \\ &= 1 - P(Z \leq 2) \\ &= 1 - 0.97725(FromStandardNormalTable[Figure\ 5.2]) \\ &= 0.02275 \end{aligned}$$

(ii) Probability that the computer fails to work for less than or equal to 2.5 hours is,

$$\begin{aligned} P(X \leq 2.5) &= P(\frac{x-\mu}{\sigma} \leq \frac{2.5-0.5}{0.75}) \\ &= P(Z \leq 2.67) \\ &= 0.9962(FromStandardNormalTable[Figure\ 5.2]) \end{aligned}$$

Example 2: The life length (in days) of electric bulb follows $N(400, 5000)$. Find the probability that the life length of a randomly selected bulb is (i) less than 450 days, (ii) more than 350 days, (iii) between 300 to 500 days.

Solution: Let, X = life length. $X \sim N(400, 5000); \mu = 400, \sigma^2 = 5000$ and $\sigma = 70.71$.

$$\begin{aligned} (i)P(X < 450) &= P(\frac{x-\mu}{\sigma} < \frac{450-400}{70.71}) \\ &= P(Z < 0.71) \\ &= 0.76115(FromStandardNormalTable[Figure\ 5.2]) \end{aligned}$$

$$\begin{aligned} (ii)P(X > 350) &= P(\frac{x-\mu}{\sigma} > \frac{350-400}{70.71}) \\ &= P(Z > -0.71) \\ &= 1 - P(Z < -0.71) \\ &= 1 - 0.23885(FromStandardNormalTable[Figure\ 5.1]) \\ &= 0.76115 \end{aligned}$$

$$\begin{aligned} (iii)P(300 < X < 500) &= P(\frac{300-400}{70.71} < \frac{x-\mu}{\sigma} < \frac{500-400}{70.71}) \\ &= P(-1.41 < Z < 1.41) \\ &= P(Z < 1.41) - P(Z < -1.41) \\ &= .92073 - .07927(FromStandardNormalTable[Figure\ 5.2\&\ 5.1respectively]) \\ &= 0.84146 \end{aligned}$$

5.2 Exponential distribution:

If X is an exponential random variable, then the probability density function is given by

$$f(x) = \frac{1}{\lambda} e^{-\frac{x}{\lambda}}; x > 0, \lambda > 0$$

Here, x = service time/length of time/waiting time, λ = average service time/length of time/ waiting time.

Examples:

- Time needed to send an email from a cyber cafe.
- Time needed to receive a signal.
- Time needed to get a service in a bank.
- Time needed to repair a mobile phone in a repairing shop.

5.2.1 Mean & variance calculation using generating functions

The moment generating function(mgf) is

$$\begin{aligned} M_x(t) = E[e^{tX}] &= \frac{1}{\lambda}\int_0^\infty e^{tx}e^{-\frac{x}{\lambda}}dx \\ &= \frac{1}{\lambda}\int_0^\infty e^{-x(\frac{1}{\lambda}-t)}dx \\ &= \frac{1}{\lambda}\int_0^\infty e^{-x(\frac{1-\lambda t}{\lambda})}dx \\ &= (1-\lambda t)^{-1} \\ &= 1+\lambda t+(\lambda t)^2+(\lambda t)^3+\cdots+(\lambda t)^\infty \\ &= \sum_{r=0}^{\infty}(\lambda t)^r \end{aligned}$$

Here, $\mu'_r = E[X^r]$ =coefficient of $\frac{t^r}{r!}$ in $M_x(t)$.

Therefore, $\mu'_1 = \lambda$, $\mu'_2 = 2\lambda^2$

Hence, Mean, μ=$\mu'_1 = \lambda$

Variance, $\mu_2 = \mu'_2 - (\mu'_1)^2 = 2\lambda^2 - \lambda^2 = \lambda^2$

Hence, $X \sim exp(\lambda, \lambda^2)$.

5.2.2 Important term for Maths

- $P(X > x) = e^{-\frac{x}{\lambda}}$
- $P(X < x) = 1 - e^{-\frac{x}{\lambda}}$
- $P(x_1 < X < x_2) = e^{-\frac{x_1}{\lambda}} - e^{-\frac{x_2}{\lambda}}$

5.2.3 Example of Exponential Distribution

Problem: The average time needed to open a computer is 2 minutes.

(a) Find the probability that a computer will be opened (i) after 3 minutes, (ii) before 2 minutes, (iii) within 2 to 3 minutes.

(b) If a man is in queue for half an hour, what is the probability that he will be able to open the computer (iv) after 35 minutes, (v) before 35 minutes, (vi) within 32 to 35 minutes.

Solution: Let, X be the time needed to open a computer. Given, $\lambda = 2$.

a(i) $P(X > 3) = e^{-\frac{x}{\lambda}} = e^{-\frac{3}{2}} = 0.2231$

a(ii) $P(X < 2) = 1 - e^{-\frac{x}{\lambda}} = 1 - e^{-\frac{2}{2}} = 0.63212$

a(iii) $P(2 < X < 3) = e^{-\frac{x_1}{\lambda}} - e^{-\frac{x_2}{\lambda}} = e^{-\frac{2}{2}} - e^{-\frac{3}{2}} = 0.14478$

b(iv) As the person is in queue for 30 minutes to open the computer, he will open the computer after $35 - 30 = 5$ minutes.
$P(X > 5) = e^{-\frac{x}{\lambda}} = e^{-\frac{5}{2}} = 0.08208$

b(v) He will open the computer before $35 - 30 = 5$ minutes.
$P(X < 5) = 1 - e^{-\frac{x}{\lambda}} = 1 - e^{-\frac{5}{2}} = 1 - 0.08208 = 0.91792$

b(vi) He will open the computer within $32 - 30 = 2$ minutes to $35 - 30 = 5$ minutes.
$P(2 < X < 5) = e^{-\frac{x_1}{\lambda}} - e^{-\frac{x_2}{\lambda}} = e^{-\frac{2}{2}} - e^{-\frac{5}{2}} = 0.2858$

5.3 Rayleigh distribution:

If X is a Rayleigh variable, then the probability density function is given by

$$f(x) = \frac{x}{\sigma^2} e^{-\frac{x^2}{2\sigma^2}}; x > 0, \sigma > 0,$$

where σ = Mode of the distribution and it is given by $\sigma = \frac{mean}{1.253} = \frac{\bar{x}}{1.253}$

Examples: Wind speed, Power of signals etc.

5.3.1 Important term for Maths

- $P(X > x) = e^{-\frac{x^2}{2\sigma^2}}$
- $P(X < x) = 1 - e^{-\frac{x^2}{2\sigma^2}}$
- $P(x_1 < X < x_2) = e^{-\frac{x_1^2}{2\sigma^2}} - e^{-\frac{x_2^2}{2\sigma^2}}$

5.3.2 Examples of Rayleigh Distribution

Example 1: The average wind speed of a day is 4.5(knot). Find the probability that in a randomly selected day, the wind speed (i) will exceed 4 knots, (ii) will be less than 3, (iii) will be between 2 to 5 knots.

Solution: Let, X be the wind speed. Given, $\bar{x} = 4.5$; then $\sigma = \frac{\bar{x}}{1.253} = \frac{4.5}{1.253} = 3.59$

(i) $P(X > 4) = e^{-\frac{x^2}{2\sigma^2}} = e^{-\frac{4^2}{2(3.59)^2}} = 0.5376$

(ii) $P(X < 3) = 1 - e^{-\frac{x^2}{2\sigma^2}} = 1 - e^{-\frac{3^2}{2(3.59)^2}} = 0.2947$

(iii) $P(2 < X < 5) = e^{-\frac{x_1^2}{2\sigma^2}} - e^{-\frac{x_2^2}{2\sigma^2}} = e^{-\frac{2^2}{2(3.59)^2}} - e^{-\frac{5^2}{2(3.59)^2}} = 0.4772$

Example 2: The mode of the density of faded out signal is 0.5. Find the probability that the density will be (i) more than 0.8, (ii) less than 0.4, (iii) between 0.4 to 0.6.

Solution: Let, X be the density of faded out signal. Given, $\sigma = 0.5$.

(i) $P(X > 0.8) = e^{-\frac{x^2}{2\sigma^2}} = e^{-\frac{0.8^2}{2(0.5)^2}} = 0.278$

(ii) $P(X < 0.4) = 1 - e^{-\frac{x^2}{2\sigma^2}} = 1 - e^{-\frac{0.4^2}{2(0.5)^2}} = 0.2739$

(iii) $P(0.4 < X < 0.6) = e^{-\frac{x_1^2}{2\sigma^2}} - e^{-\frac{x_2^2}{2\sigma^2}} = e^{-\frac{0.4^2}{2(0.5)^2}} - e^{-\frac{0.6^2}{2(0.5)^2}} = 0.2393$

5.4 Exercise:

1. The time (in minutes) to write a program follows $N(30, 505)$. Find the probability that a program will be written (i) after 25 minutes, (ii) before 30 minutes, (iii) between 20 to 25 minutes.
2. The consumption of electricity (in MW) per day in a city follows $N(250, 3600)$. Find the probability that in a randomly selected day the consumption will (i) exceed 260, (ii) be less than 200, (iii) be 225 t0 275.
3. The average time needed to get an internet connection is 5 minutes. If a man is in queue for 15 minutes, what is the probability that he will get connection- (i) before 18 minutes, (ii) after 20 minutes, (iii) between 17 to 20 minutes?
4. The average power (x watts) of signal is 3. Find the probability that the power of the signal will be (i) less than 2, (ii) between 2.5 and 3.5, (iii) more than 3.

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.9	.00005	.00005	.00004	.00004	.00004	.00004	.00004	.00004	.00003	.00003
-3.8	.00007	.00007	.00007	.00006	.00006	.00006	.00006	.00005	.00005	.00005
-3.7	.00011	.00010	.00010	.00010	.00009	.00009	.00008	.00008	.00008	.00008
-3.6	.00016	.00015	.00015	.00014	.00014	.00013	.00013	.00012	.00012	.00011
-3.5	.00023	.00022	.00022	.00021	.00020	.00019	.00019	.00018	.00017	.00017
-3.4	.00034	.00032	.00031	.00030	.00029	.00028	.00027	.00026	.00025	.00024
-3.3	.00048	.00047	.00045	.00043	.00042	.00040	.00039	.00038	.00036	.00035
-3.2	.00069	.00066	.00064	.00062	.00060	.00058	.00056	.00054	.00052	.00050
-3.1	.00097	.00094	.00090	.00087	.00084	.00082	.00079	.00076	.00074	.00071
-3.0	.00135	.00131	.00126	.00122	.00118	.00114	.00111	.00107	.00104	.00100
-2.9	.00187	.00181	.00175	.00169	.00164	.00159	.00154	.00149	.00144	.00139
-2.8	.00256	.00248	.00240	.00233	.00226	.00219	.00212	.00205	.00199	.00193
-2.7	.00347	.00336	.00326	.00317	.00307	.00298	.00289	.00280	.00272	.00264
-2.6	.00466	.00453	.00440	.00427	.00415	.00402	.00391	.00379	.00368	.00357
-2.5	.00621	.00604	.00587	.00570	.00554	.00539	.00523	.00508	.00494	.00480
-2.4	.00820	.00798	.00776	.00755	.00734	.00714	.00695	.00676	.00657	.00639
-2.3	.01072	.01044	.01017	.00990	.00964	.00939	.00914	.00889	.00866	.00842
-2.2	.01390	.01355	.01321	.01287	.01255	.01222	.01191	.01160	.01130	.01101
-2.1	.01786	.01743	.01700	.01659	.01618	.01578	.01539	.01500	.01463	.01426
-2.0	.02275	.02222	.02169	.02118	.02068	.02018	.01970	.01923	.01876	.01831
-1.9	.02872	.02807	.02743	.02680	.02619	.02559	.02500	.02442	.02385	.02330
-1.8	.03593	.03515	.03438	.03362	.03288	.03216	.03144	.03074	.03005	.02938
-1.7	.04457	.04363	.04272	.04182	.04093	.04006	.03920	.03836	.03754	.03673
-1.6	.05480	.05370	.05262	.05155	.05050	.04947	.04846	.04746	.04648	.04551
-1.5	.06681	.06552	.06426	.06301	.06178	.06057	.05938	.05821	.05705	.05592
-1.4	.08076	.07927	.07780	.07636	.07493	.07353	.07215	.07078	.06944	.06811
-1.3	.09680	.09510	.09342	.09176	.09012	.08851	.08691	.08534	.08379	.08226
-1.2	.11507	.11314	.11123	.10935	.10749	.10565	.10383	.10204	.10027	.09853
-1.1	.13567	.13350	.13136	.12924	.12714	.12507	.12302	.12100	.11900	.11702
-1.0	.15866	.15625	.15386	.15151	.14917	.14686	.14457	.14231	.14007	.13786
-0.9	.18406	.18141	.17879	.17619	.17361	.17106	.16853	.16602	.16354	.16109
-0.8	.21186	.20897	.20611	.20327	.20045	.19766	.19489	.19215	.18943	.18673
-0.7	.24196	.23885	.23576	.23270	.22965	.22663	.22363	.22065	.21770	.21476
-0.6	.27425	.27093	.26763	.26435	.26109	.25785	.25463	.25143	.24825	.24510
-0.5	.30854	.30503	.30153	.29806	.29460	.29116	.28774	.28434	.28096	.27760
-0.4	.34458	.34090	.33724	.33360	.32997	.32636	.32276	.31918	.31561	.31207
-0.3	.38209	.37828	.37448	.37070	.36693	.36317	.35942	.35569	.35197	.34827
-0.2	.42074	.41683	.41294	.40905	.40517	.40129	.39743	.39358	.38974	.38591
-0.1	.46017	.45620	.45224	.44828	.44433	.44038	.43644	.43251	.42858	.42465
-0.0	.50000	.49601	.49202	.48803	.48405	.48006	.47608	.47210	.46812	.46414

Figure 5.1: Standard Normal Table

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790	.53188	.53586
0.1	.53983	.54380	.54776	.55172	.55567	.55962	.56356	.56749	.57142	.57535
0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642	.61026	.61409
0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891
1.0	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769	.85993	.86214
1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900	.88100	.88298
1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796	.89973	.90147
1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91309	.91466	.91621	.91774
1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922	.93056	.93189
1.5	.93319	.93448	.93574	.93699	.93822	.93943	.94062	.94179	.94295	.94408
1.6	.94520	.94630	.94738	.94845	.94950	.95053	.95154	.95254	.95352	.95449
1.7	.95543	.95637	.95728	.95818	.95907	.95994	.96080	.96164	.96246	.96327
1.8	.96407	.96485	.96562	.96638	.96712	.96784	.96856	.96926	.96995	.97062
1.9	.97128	.97193	.97257	.97320	.97381	.97441	.97500	.97558	.97615	.97670
2.0	.97725	.97778	.97831	.97882	.97932	.97982	.98030	.98077	.98124	.98169
2.1	.98214	.98257	.98300	.98341	.98382	.98422	.98461	.98500	.98537	.98574
2.2	.98610	.98645	.98679	.98713	.98745	.98778	.98809	.98840	.98870	.98899
2.3	.98928	.98956	.98983	.99010	.99036	.99061	.99086	.99111	.99134	.99158
2.4	.99180	.99202	.99224	.99245	.99266	.99286	.99305	.99324	.99343	.99361
2.5	.99379	.99396	.99413	.99430	.99446	.99461	.99477	.99492	.99506	.99520
2.6	.99534	.99547	.99560	.99573	.99585	.99598	.99609	.99621	.99632	.99643
2.7	.99653	.99664	.99674	.99683	.99693	.99702	.99711	.99720	.99728	.99736
2.8	.99744	.99752	.99760	.99767	.99774	.99781	.99788	.99795	.99801	.99807
2.9	.99813	.99819	.99825	.99831	.99836	.99841	.99846	.99851	.99856	.99861
3.0	.99865	.99869	.99874	.99878	.99882	.99886	.99889	.99893	.99896	.99900
3.1	.99903	.99906	.99910	.99913	.99916	.99918	.99921	.99924	.99926	.99929
3.2	.99931	.99934	.99936	.99938	.99940	.99942	.99944	.99946	.99948	.99950
3.3	.99952	.99953	.99955	.99957	.99958	.99960	.99961	.99962	.99964	.99965
3.4	.99966	.99968	.99969	.99970	.99971	.99972	.99973	.99974	.99975	.99976
3.5	.99977	.99978	.99978	.99979	.99980	.99981	.99981	.99982	.99983	.99983
3.6	.99984	.99985	.99985	.99986	.99986	.99987	.99987	.99988	.99988	.99989
3.7	.99989	.99990	.99990	.99990	.99991	.99991	.99992	.99992	.99992	.99992
3.8	.99993	.99993	.99993	.99994	.99994	.99994	.99994	.99995	.99995	.99995
3.9	.99995	.99995	.99996	.99996	.99996	.99996	.99996	.99996	.99997	.99997

Figure 5.2: Standard Normal Table